

Question number	Answer	Mark
8(a)(i)	B 2	(1)

Question number	Answer	Mark
8(a)(ii)	An explanation that includes the following points: • glucose increases as glycogen decreases (1) • because glycogen is being hydrolysed (1)	(2)

Question number	Answer	Mark
8(b)(i)	• calculation of percentage difference Example of calculation: $((106 - 28) \div 28) \times 100 = 279 / 278.6 / 278.57$	(1)

Question number	Answer	Mark
8(b)(ii)	An explanation that includes the following points: • data from the wood frog from Alaska are more reliable as standard deviation is relatively smaller (1) • 9.43% compared to 17.86% of mean (1)	(2)

Question number	Answer	Mark
8(b)(iii)	An explanation that includes the following points: • solutes lower the freezing point of the plasma (1) • prevents damage to the cells by {freezing / ice crystals} (1)	(2)

Question number	Answer	Mark
8(c)	An explanation that includes any five of the following points: • natural selection (1) • because a mutation in DNA occurred (1) • resulting in alleles for increased solutes in plasma (1) • therefore these wood frogs could survive in colder areas (1) • reducing competition with other wood frogs (1) • therefore survived and reproduced passing on these alleles (1)	(5)

